

Concepts & Examples: Lagrange, Divided Difference, and Cubic Spline Interpolations

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Intro & Similarities

We will be discussing some 1-D interpolation methods with some pros and cons:

Lagrange interpolation is suitable for small datasets, where analytically the interpolant function can closely mimic the true, unknown function.

Newton's DD is an extension to the method above to increase precision. This works well with more unstructured datasets to iteratively compute parameters.

Cubic splines is more applicable to continuous dataset with higher uncertainty. Uses pairwise splines to map plausible function values at each interval.

Downsides to these techniques is that it doesn't make good predictions on large datasets as these methods are heavily data dependent – Only taking into account subtle magnitude & gradient differences.

Lagrange Interpolation

The polynomial is constructed through a linear combination of Kronecker Delta functions with the function values at each point.

Assume you have a vector of your x values, the Kronecker Delta function is defined by a quotient in respect to the dependent variable at the function node. This quotient is a ratio on the product of the differences between non-target x values and target x value. Given this is a function, only the denominator holds a numeric value while the numerator is in terms of the dependent variable at node.

Examples: estimating population census data, tracking athletes cartesian coordinates

Given points:

$$(0, 1), \quad (1, 0.5), \quad (3, 4)$$

The Lagrange polynomial is:

$$P_2(x) = \sum_{i=0}^3 y_i L_i(x)$$

$$L_0(x) = \frac{(x-1)(x-3)}{3}, \quad L_1(x) = \frac{x(x-3)}{-2}, \quad L_2(x) = \frac{x(x-1)}{6}$$

$$P_2(x) = \frac{(x-1)(x-3)}{3} - \frac{0.5 \cdot x(x-3)}{2} + \frac{4 \cdot x(x-1)}{6}$$

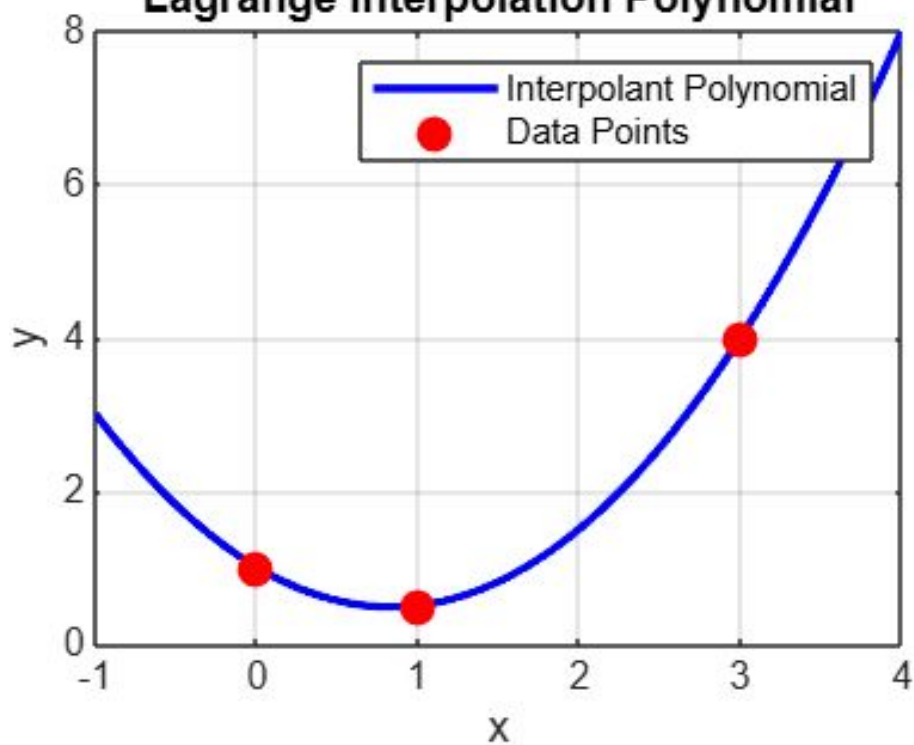
Expanding terms:

$$P_2(x) = \frac{x^2 - 4x + 3}{3} - \frac{x^2 - 3x}{4} + \frac{2x^2 - 2x}{3}$$

Combining like terms:

$$P_2(x) = \frac{3}{4}x^2 - \frac{5}{4}x + 1$$

Lagrange Interpolation Polynomial



Newton's Divided Difference

Divided differences is a more structured and recursive way to compute polynomials when getting additional data points.

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n(x - x_0) \cdots (x - x_{n-1})$$

Evaluating at x_0 and x_1 give us the following terms which we can rewrite in divided difference notation.

$$a_0 = P_n(x_0) = f(x_0).$$

$$f[x_i] = f(x_i).$$

$$f(x_0) + a_1(x_1 - x_0) = P_n(x_1) = f(x_1);$$

$$a_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}.$$

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}.$$

Divided Differences

nth divided difference
$$f[x_0, x_1, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}.$$

x	$f(x)$	First divided differences	Second divided differences	Third divided differences
x_0	$f[x_0]$			
		$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$		
x_1	$f[x_1]$		$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	
		$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$		$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$
x_2	$f[x_2]$		$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	
		$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2}$		$f[x_1, x_2, x_3, x_4] = \frac{f[x_2, x_3, x_4] - f[x_1, x_2, x_3]}{x_4 - x_1}$
x_3	$f[x_3]$		$f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{x_4 - x_2}$	
		$f[x_3, x_4] = \frac{f[x_4] - f[x_3]}{x_4 - x_3}$		$f[x_2, x_3, x_4, x_5] = \frac{f[x_3, x_4, x_5] - f[x_2, x_3, x_4]}{x_5 - x_2}$
x_4	$f[x_4]$		$f[x_3, x_4, x_5] = \frac{f[x_4, x_5] - f[x_3, x_4]}{x_5 - x_3}$	
		$f[x_4, x_5] = \frac{f[x_5] - f[x_4]}{x_5 - x_4}$		
x_5	$f[x_5]$			

Divided Difference (ex.)

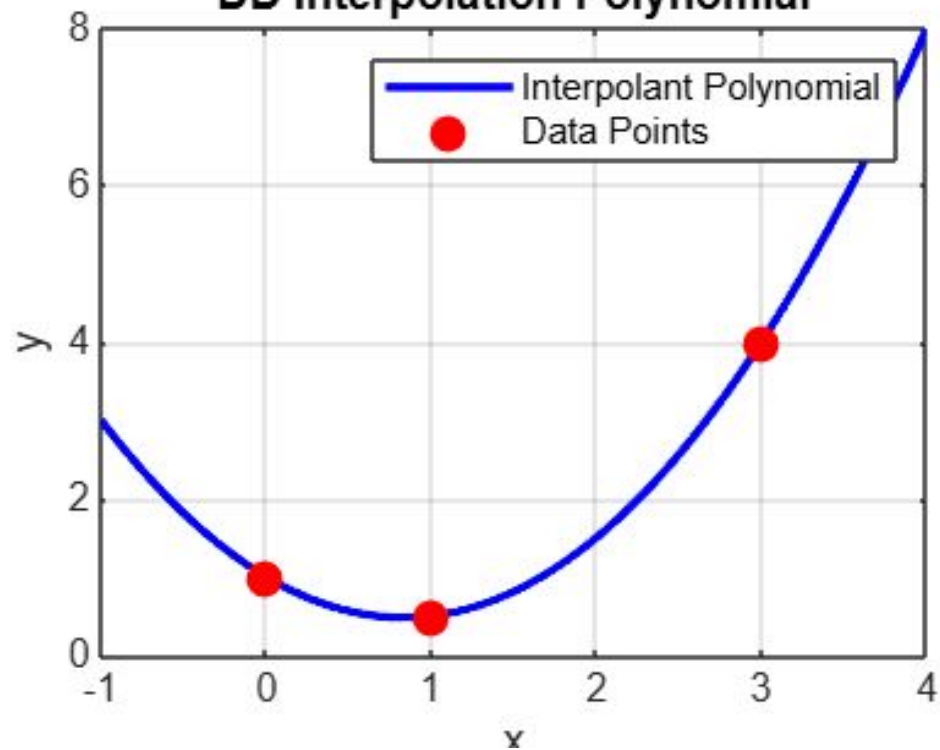
(0,1), (1,0.5), (3,4)

$$P_2 = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1)$$

x	f(x)	1st DD	2nd DD
0	1	-0.5	0.75
1	0.5	1.75	
3	4		

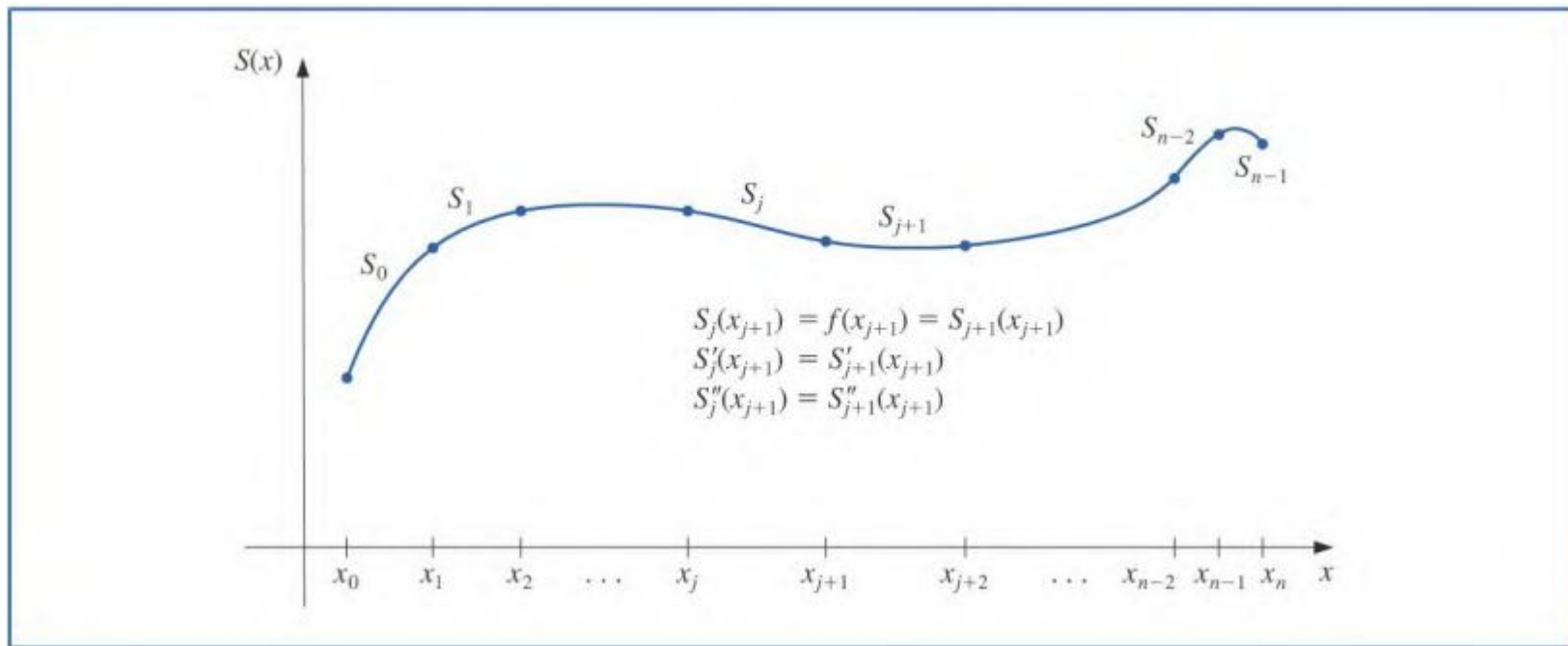
$$\begin{aligned}P_2(x) &= 1 - 0.5(x - x_0) + 0.75(x - x_0)(x - x_1) \\&= 1 - 0.5x + 0.75x(x - 1) \\&= \frac{3}{4}x^2 - \frac{5}{4}x + 1\end{aligned}$$

DD Interpolation Polynomial



Cubic Splines

Given a set of data points, a cubic spline interpolant is a piecewise cubic function that passes through the given points while maintaining continuous first and second derivatives.



Cubic Splines

Given a function defined on $[a, b]$ and a set of nodes $a = x_0 < x_1 < \dots < x_n = b$, a cubic spline interpolant S for f is a function that satisfies the following conditions:

- (a) $S(x)$ is a cubic polynomial, denoted $S_j(x)$, on the subinterval $[x_j, x_{j+1}]$ for each $j = 0, 1, \dots, n-1$;
- (b) $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1})$ for each $j = 0, 1, \dots, n-1$;
- (c) $S_{j+1}(x_{j+1}) = S_j(x_{j+1})$ for each $j = 0, 1, \dots, n-2$; (*Implied by (b).*)
- (d) $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$ for each $j = 0, 1, \dots, n-2$;
- (e) $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$ for each $j = 0, 1, \dots, n-2$;
- (f) One of the following sets of boundary conditions is satisfied:
 - (i) $S''(x_0) = S''(x_n) = 0$ (**natural (or free) boundary**);
 - (ii) $S'(x_0) = f'(x_0)$ and $S'(x_n) = f'(x_n)$ (**clamped boundary**).

Cubic Spline Interpolation for $(0, 1)$, $(1, 0.5)$, $(3, 4)$

Define

$$S_0(x) = a_0 + b_0(x - 0) + c_0(x - 0)^2 + d_0(x - 0)^3, \quad x \in [0, 1],$$

$$S_1(x) = a_1 + b_1(x - 1) + c_1(x - 1)^2 + d_1(x - 1)^3, \quad x \in [1, 3].$$

Interpolation Conditions:

$$(1) \quad S_0(0) = a_0 = 1,$$

$$(2) \quad S_0(1) = a_0 + b_0 + c_0 + d_0 = 0.5,$$

$$(3) \quad S_1(1) = a_1 = 0.5,$$

$$(4) \quad S_1(3) = a_1 + 2b_1 + 4c_1 + 8d_1 = 4.$$

Continuity Conditions at $x = 1$:

$$(5) \quad S'_0(1) = b_0 + 2c_0 + 3d_0 = S'_1(1) = b_1,$$

$$(6) \quad S''_0(1) = 2c_0 + 6d_0 = S''_1(1) = 2c_1.$$

Natural Spline Boundary Conditions:

$$(7) \quad S''_0(0) = 2c_0 = 0,$$

$$(8) \quad S''_1(3) = 2c_1 + 12d_1 = 0.$$

$$S_0(x) = 1 - 0.875x + 0.3750x^3$$

$$S_1(x) = 0.5 + 0.25(x - 1) + 1.125(x - 1)^2 - 0.1875(x - 1)^3$$

Cubic Spline Interpolation

